

ZytoLight® SPEC MYC/IGH Dual Color Dual Fusion Probe**Intended Purpose**

The ZytoLight SPEC MYC/IGH Dual Color Dual Fusion Probe (PL62) is intended to be used for the qualitative detection of the translocation t(8;14)(q24.21;q32.3) involving the human IGH and MYC genes in formalin-fixed, paraffin-embedded specimens by fluorescence *in situ* hybridization (FISH). The probe is intended to be used in combination with the ZytoLight FISH-Tissue Implementation Kit (Prod. No. Z-2028-5/-20).

The product is intended for professional use only. All tests using the product should be performed in a certified, licensed anatomic pathology laboratory under the supervision of a pathologist/human geneticist by qualified personnel.

The probe is intended to be used as an aid to the differential diagnosis of various cancers and therapeutic measures should not be initiated based on the test result alone.

Probe Description

The ZytoLight SPEC MYC/IGH Dual Color Dual Fusion Probe is composed of:

- ZyGreen (excitation 503 nm/emission 528 nm) labeled polynucleotides (~12 ng/μl), which target sequences mapping in 14q32.33* (chr14:105,462,169-106,995,000) harboring the IGH locus (see Fig. 1).
- ZyOrange (excitation 547 nm/emission 572 nm) labeled polynucleotides (~6 ng/μl), which target sequences mapping in 8q24.21* (chr8:128,171,178-129,517,468) harboring the MYC gene region (see Fig. 1).
- Formamide based hybridization buffer

* according to Human Genome Assembly GRCh37/hg19

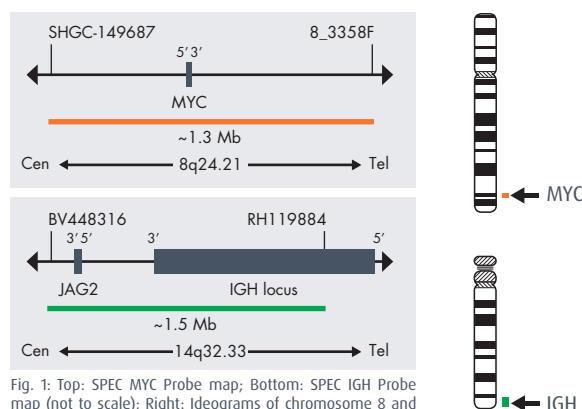
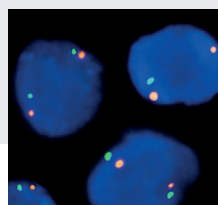


Fig. 1: Top: SPEC MYC Probe map; Bottom: SPEC IGH Probe map (not to scale); Right: Ideograms of chromosome 8 and 14 indicating the hybridization locations

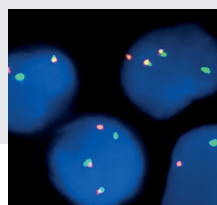
Results

Normal situation: In interphases of normal cells or cells without a translocation involving the respective bands, two separate green/orange signals appear.

Aberrant situation: A gene fusion is indicated by one separate orange signal, one separate green signal, and two orange/green fusion signals.



Hybridized to normal interphase cells as indicated by two orange and two green signals in each nucleus.



Example of an aberrant signal pattern: Burkitt lymphoma tissue section with t(8;14) as indicated by one separate orange signal, one separate green signal and two orange/green fusion signals indicating the MYC/IGH translocation.

Labels	Chromosome	Reg. Status	Tests	Order No.
	8, 14	CE/IVD	5 (50 μl)	Z-2105-50
			20 (200 μl)	Z-2105-200

ZytoLight® SPEC CD274,PDCD1LG2/CEN 9 Dual Color Probe**Intended Purpose**

The ZytoLight SPEC CD274,PDCD1LG2/CEN 9 Dual Color Probe (PL138) is intended to be used for the qualitative detection of amplifications involving the human CD274,PDCD1LG2 gene cluster as well as the detection of the classical satellite III region of chromosome 9 in formalin-fixed, paraffin-embedded specimens by fluorescence *in situ* hybridization (FISH). The probe is intended to be used in combination with the ZytoLight FISH-Tissue Implementation Kit (Prod. No. Z-2028-5/-20).

The product is intended for professional use only. All tests using the product should be performed in a certified, licensed anatomic pathology laboratory under the supervision of a pathologist/human geneticist by qualified personnel.

The probe is intended to be used as an aid to the differential diagnosis of various cancers and therapeutic measures should not be initiated based on the test result alone.

Probe Description

The ZytoLight SPEC CD274,PDCD1LG2/CEN 9 Dual Color Probe is composed of:

- ZyGreen (excitation 503 nm/emission 528 nm) labeled polynucleotides (~10 ng/μl), which target sequences mapping in 9p24.1* (chr9:5,253,553-5,819,972) harboring the CD274, PDCD1LG2 gene cluster (see Fig. 1).
- ZyOrange (excitation 547 nm/emission 572 nm) labeled polynucleotides (~1.5 ng/μl), which target sequences mapping in 9q12 specific for the classical satellite III centromeric region D9Z3 of chromosome 9.
- Formamide based hybridization buffer

* according to Human Genome Assembly GRCh37/hg19

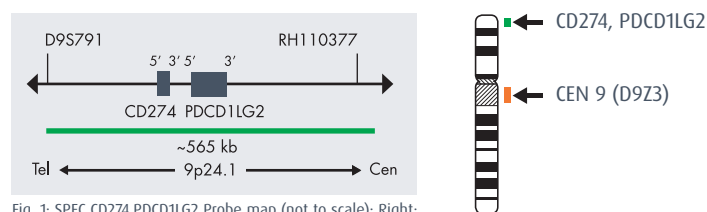
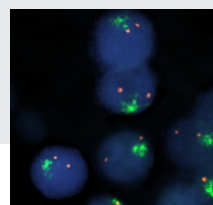


Fig. 1: SPEC CD274,PDCD1LG2 Probe map (not to scale); Right: Ideogram of chromosome 9 indicating the hybridization locations

Results

Normal situation: In interphases of normal cells or cells without an amplification involving the CD274,PDCD1LG2 gene cluster, two green and two orange signals appear.

Aberrant situation: In cells with an amplification of the CD274,PDCD1LG2 gene cluster, an increased number of green signals or green signal clusters will be observed.



Example of an aberrant signal pattern: Primary mediastinal large B-cell lymphoma tissue section with amplification of the CD274,PDCD1LG2 gene region as indicated by green signal clusters in each nucleus.

Labels	Chromosome	Reg. Status	Tests	Order No.
	9	CE/IVD	5 (50 μl)	Z-2179-50
			20 (200 μl)	Z-2179-200